

Luteinizing Hormone (LH)

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

The primary clinical use of LH measurement is in evaluating the normalcy of hypothalamic-pituitary-gonadal axis. Measurement of serum gonadotropin levels will allow for distinguishing between primary gonadal failure and deficient gonadal stimulation. LH measurement may also be of clinical importance because growth hormone and LH are frequently the first hormones to be affected by pituitary disease. The serum analysis of LH has also been found to be very useful in the diagnosis and treatment of infertility in women.

Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

Secretion of both LH and FSH are pulsatile, in response to the normal intermittent release of gonadotropin-releasing hormone (GnRH). In addition, in females, both FSH and LH vary over the course of the menstrual cycle, with peaks at time of ovulation. Thus, interpretation of a single determination may be difficult. It has been suggested that samples be obtained at 15- to 30-minute intervals and equal volumes of serum be pooled to decrease the effect of pulsatile secretion.

As with all tests containing monoclonal mouse antibodies, erroneous findings may be obtained from samples taken from patients who have been treated with monoclonal mouse antibodies or who have received them for diagnostic purposes.¹ In rare cases, interference due to extremely high titers of antibodies to streptavidin and ruthenium can occur.¹ The test contains additives, which minimize these effects.

Custom Additional Information

LH (luteinizing hormone), together with FSH (follicle-stimulating hormone), belongs to the gonadotropin family. LH and FSH regulate and stimulate the growth and function of the gonads (ovaries and testes) synergistically.²⁻⁴ Like FSH, TSH, and hCG, LH is a glycoprotein consisting of two subunits (α - and β -chains). This proteohormone, which consists of 121 amino acids³ and three sugar chains, has a molecular weight of 29,500 daltons.⁴

In women, the gonadotropins act within the hypothalamus-pituitary-ovary regulating circuit to control the menstrual cycle.^{5,6} LH and FSH are released in pulses from the gonadotropic cells of the anterior pituitary and pass via the bloodstream to the ovaries. Here the gonadotropins stimulate the growth and maturation of the follicle and hence the biosynthesis of estrogens and progesterones. The highest LH-concentrations occur during the midcycle peak and induce ovulation and formation of the corpus luteum, the principal secretion product of which is progesterone.

Determination of LH concentration is used in the elucidation of dysfunctions within the hypothalamus-pituitary-gonads system. In the Leydig cells of the testes, LH stimulates the production of testosterone.

The determination of LH in conjunction with FSH is utilized for the following indications: congenital diseases with chromosome aberrations (eg, Turner syndrome), polycystic ovaries (PCO), clarifying the causes of amenorrhea, menopausal syndrome, and suspected Leydig cell insufficiency.^{2,5,6}

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

If a red-top tube or plasma is used, transfer separated serum or plasma to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.

Pediatric (Male and Female) (mIU/mL)		
<24 h	<0.2–1.0	
1 d	<0.2–0.8	
2 d	<0.2–0.6	
3 d	<0.2–2.7	
4 d	<0.2–1.7	
5 d	<0.2–3.1	
6 d	0.4–6.4	
7 d	<0.2–5.6	
8 to 30 d	<0.2–7.8	
Age	Pediatric (Male)	Pediatric (Female)
1 to 12 m	<0.2–0.4	<0.2–0.4
1 to 4 y	<0.2–1.3	<0.2–0.5
5 to 9 y	<0.2–1.4	<0.2–3.1
10 to 12 y	<0.2–7.8	<0.2–11.9
13 to 16 y	1.3–9.8	0.5–41.7
Adult Male (mIU/mL): 1.7–8.6		
Adult Female (mIU/mL)		

Pediatric (Male and Female) (mIU/mL)	
follicular	2.4–12.6
ovulation	14.0–95.6
luteal	1.0–11.4
postmenopausal	7.7–58.5

Storage Instructions

Room temperature

References

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Footnotes

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6. Scott MG, Ladenson JH, Green ED, Gast MJ. Hormonal evaluation of female infertility and reproductive disorders. *Clin Chem*. 1989 Apr; 35(4):620-630. PubMed 2522836 (<http://www.ncbi.nlm.nih.gov/pubmed/2522836>)